

Module 2 Assignment: Formulating a Design Plan

585.617 Rehabilitation Engineering / Module 2 Assignment / [Your Name] / [Date]

Design Prompt: Retractable Sun and Rain Awning for a Powered Wheelchair

Part A — Need Statement

Need statement: *A way for powered wheelchair users to obtain on-demand overhead protection from sun and precipitation, deployable and stowable independently from the seated position, without compromising the chair's stability, its dimensional envelope in everyday indoor and vehicle-transport environments, or its range.*

Supporting rationale. The target population is full-time powered wheelchair users. Two characteristics of this population make weather exposure a functional problem rather than a comfort problem. First, individuals with spinal cord injury at or above roughly T6 lose sympathetic control of sudomotor and vasomotor function below the lesion, so they cannot dissipate heat by sweating over most of the body surface and are at elevated risk of hyperthermia under sustained sun exposure [4]. Second, the joystick, controller, and connector housings are the most moisture-vulnerable subsystem of a powered chair, so users routinely forgo travel when rain is forecast. In ICF terms the result is a *participation restriction* produced by an environmental factor rather than by the underlying impairment — precisely the class of problem rehabilitation engineering exists to address. Users also cannot reposition into shade or shelter as quickly as ambulatory people can, and many have limited hand function, so any solution requiring a caregiver to erect it has already failed the need.

Note on solution neutrality. The task as assigned (“add a retractable awning”) specifies a solution, not a need. Following Dieter and Schmid [3], the statement above is written solution-neutrally so that concept generation is not foreclosed before the design input phase: a canopy is one embodiment, but a deployable hood, an integrated garment, or a controller-enclosure upgrade paired with weather alerting may screen better against the same needs. The plan below carries the retractable awning as the presumptive concept while preserving a genuine concept-screening gate in Phase 2.

Part B — Design Plan (FDA Waterfall)

This plan follows the design control waterfall of the FDA *Design Control Guidance for Medical Device Manufacturers* [1], as required by 21 CFR 820.30 [2]. Because the awning is an accessory to an already-marketed Class II powered wheelchair, the project is executed as a design change carried through the full waterfall, and a change assessment under FDA’s 510(k)-change guidance [11] is an explicit Phase 1 deliverable. All outputs are filed in the Design History File (DHF), and every design review includes at least one individual without direct responsibility for the stage under review, per 21 CFR 820.30(e).

Critical User Needs and Traced Design Inputs

ID	Critical user need	ID	Critical design input (objective and verifiable)
UN-1	The user must be able to deploy and stow the awning without assistance, from the seated position, with impaired hand function and without reaching overhead.	DI-1.1	A full deploy or stow cycle shall complete in ≤ 10 s, initiated by a single control requiring ≤ 22 N (5 lbf) of force and no tight grasping, pinching, or twisting of the wrist (ADA §309.4 [6]).
		DI-1.2	The control shall sit within the seated forward and side reach envelope, require no reach above shoulder height, and be operable single-handed by a user with C6 tetraplegia (tenodesis grasp, no active finger flexion).
UN-2	The awning must not prevent the user from going where they normally go, or make the chair feel unsafe — indoors, on ramps, in wind, or during vehicle transport.	DI-2.1	Stowed, the accessory shall not alter the chair’s ISO 7176-5 overall length, width, or turning envelope. Deployed, occupied overall height shall not exceed 1500 mm and overall width shall not exceed the base chair’s width.
		DI-2.2	With the awning deployed, the chair shall meet its baseline ISO 7176-1 static and ISO 7176-2 dynamic stability limits [5], and shall auto-stow above a wind threshold established by analysis and confirmed by test.
UN-3	The awning must actually keep sun	DI-3.1	Canopy fabric shall provide $UPF \geq 50$ and hydrostatic head $\geq$

ID	Critical user need	ID	Critical design input (objective and verifiable)
	and rain off the user and off the chair's electronics.		1000 mm (ISO 811). Deployed coverage shall shade the occupant's head, shoulders, and joystick at solar elevations 30°-90°, with no loss of controller function under IPX4 spray [5].

Phase 0 — User Needs (Weeks 1-8)

Tasks: recruit and interview 10-15 powered chair users spanning diagnoses and hand function, plus 3-5 clinicians (PT/OT, ATP, seating clinic); observe use in situ; draft the need statement; build a preliminary hazard list; draft the business case. *Information to gather:* weather-avoidance frequency and consequence; caregiver availability; existing workarounds (umbrella clamps, ponchos); anthropometric and reach data for seated users; competitive and patent landscape; warranty history for moisture-related controller failures. *Design review DR-1 (Conceptual — “Should we build this?”):* project engineer, product management, clinical advisor, regulatory affairs, quality, manufacturing, independent reviewer; presenting the need statement, user research summary, competitive and patent landscape, preliminary hazard analysis, and business case. *Exit criteria:* user needs list approved and signed; every need attributable to primary research rather than assumption; no unresolved showstopper hazard; funded go decision.

Phase 1 — Design Input (Weeks 9-18)

Tasks: convert every user need into at least one objective, verifiable design input; draft the Product Design Specification; open the ISO 14971 risk management file [8]; perform the 510(k)-change assessment [11]; establish the verification strategy and the human factors plan per IEC 62366-1 and FDA HFE guidance [9]. *Information to gather:* applicable ISO 7176 parts and their acceptance limits [5]; IEC Guide 71 provisions on reach, force, and legibility for older and disabled users [7]; the base chair's battery and range budget; controller bus capability for actuator integration. *Design review DR-2 (Preliminary — “Will it work?”):* the Phase 0 attendees plus test engineering and a human factors specialist; presenting the PDS, requirements traceability matrix, risk analysis, verification strategy, and regulatory pathway determination. *Exit criteria:* 100% of user needs traced to at least one design input; every input stated in measurable terms with a named verification method; every hazard assigned a risk control; regulatory pathway documented.

Phase 2 — Design Process (Weeks 19-34)

Tasks: generate at least six concepts; screen them with a Pugh matrix against the weighted user needs; build functional prototypes of the top two; run the engineering analyses below; freeze the design. *Information to gather:* supplier actuator data, fabric test certificates, and the measured center of gravity of the loaded base chair. **Technical aspects considered at this phase —** two worked examples:

- **TA-1 — Wind loading and center-of-gravity shift.** The canopy is a sail mounted above the occupant's CG, so it adds both a lateral force and a long moment arm. Sizing to $F = \frac{1}{2}\rho v^2 C_d A$ with $A \approx 0.5 \text{ m}^2$, $C_d \approx 1.2$ and $\rho = 1.225 \text{ kg/m}^3$ gives roughly 37 N at 10 m/s (22 mph) applied near 1.4 m height — a tipping moment large enough to threaten the ISO 7176-1 stability margin. This analysis sets the auto-stow threshold in DI-2.2 and drives mast placement, so it belongs on the critical path rather than in a late design check.
- **TA-2 — Material selection under competing objectives.** Canopy ribs are traded among 6061-T6 aluminium, glass-filled PA66, and pultruded GFRP against stiffness-to-weight, UV and salt resistance, cost, and recyclability; canopy fabric is traded between solution-dyed acrylic and PU-coated polyester against UPF retention, hydrostatic head, hand feel, and packed volume. Green engineering enters as a stated screening criterion rather than an afterthought: the screen favours a mono-material, mechanically disassemblable canopy assembly and excludes PVC coatings and PFAS-based durable water repellents, accepting a modest weight penalty in exchange for end-of-life recyclability.

Anthropometric considerations run through both. Seated eye height and overhead clearance set canopy height; ADA §307.4's 80 in (2030 mm) protruding-object limit and the typical 32 in doorway clear width bound the deployed envelope [6]; accessible-van door heights bound it further; and seated forward and side reach envelopes fix control placement per DI-1.2. *Design review DR-2b (concept gate):* engineering, clinical advisor, manufacturing, and the independent

reviewer, presenting the Pugh matrix and prototype data. *Exit criteria*: one concept selected with documented rationale; the prototype demonstrates feasibility against DI-1.1, DI-2.2, and DI-3.1; no open high-severity risk lacking a control.

Phase 3 — Design Output (Weeks 31-42, overlapping)

Tasks: release drawings, specifications, BOM, actuator control firmware, labelling and IFU; complete the DFMEA; draft the Device Master Record; write and approve every V&V protocol; submit the home-use study protocol for IRB review. *Information to gather*: supplier qualification data and tooling capability studies. *Design review DR-3 (Critical — “Will it be successful?”)*: the full cross-functional team plus supplier quality and the independent reviewer, presenting the complete output package, DFMEA, essential-output identification, and approved protocols. *Exit criteria*: every design output traces to a design input; essential outputs identified for manufacturing acceptance; all verification and validation protocols approved **before** any testing begins.

Phase 4 — Verification (Weeks 43-54) — “did we design the device right?”

ID	Verifies	Verification plan
VER-1	DI-1.1, DI-1.2	Instrumented actuation test on 25 production-equivalent units: a force gauge measures peak control force and a timer measures full deploy and stow cycles across the specified temperature range. <i>Acceptance</i> : peak force ≤ 22 N and cycle time ≤ 10 s on 100% of units. Followed by a 10,000-cycle durability run built on the ISO 7176-8 fatigue philosophy [5], with force and time re-measured at completion and no loss of function permitted.
VER-2	DI-2.1, DI-2.2	Dimensional verification per ISO 7176-5 stowed and deployed, then static stability per ISO 7176-1 and dynamic stability per ISO 7176-2 with the awning deployed, across the chair’s rated occupant mass range. Lateral load is then applied by calibrated fan or wind tunnel in 2 m/s increments to confirm the auto-stow threshold actuates below the measured tipping condition, with margin. <i>Acceptance</i> : stowed envelope unchanged; deployed configuration meets baseline stability limits; auto-stow triggers at or below threshold on 20 of 20 trials.
VER-3	DI-3.1	Fabric UPF and hydrostatic head (ISO 811) on three production lots; shade-coverage mapping against a seated anthropometric manikin at solar elevations of 30°, 60°, and 90°; IPX4 spray test per IEC 60529 with the controller powered and logged; climatic conditioning per ISO 7176-9 followed by re-test.

Phase 5 — Validation (Weeks 55-68) — “did we design the right device?”

Tasks: human factors validation with 15 representative users per FDA HFE guidance [9]; a four-week in-home and community use study on production-equivalent units; residual risk-benefit determination. **Validation survey questions** — two worked examples tied to the critical user needs, with the remainder of the instrument built around QUEST 2.0 [10]:

- **VAL-Q1 (validates UN-1)**: “Over the past week, on how many occasions did you open or close the awning, and on how many of those occasions did you need another person’s help?” — followed by the five-point agreement item, “I was able to open and close the awning by myself, quickly enough to matter when the weather changed.” *Acceptance*: $\geq 90\%$ of deployments unassisted and $\geq 80\%$ of participants agreeing or strongly agreeing.
- **VAL-Q2 (validates UN-2)**: “During the study, did the awning stop you from going anywhere you would normally go — through a doorway, into an elevator, onto a ramp, or into your vehicle? If yes, describe what happened.” — paired with, “Did you at any time feel unsteady, or that the chair might tip, while the awning was open? Describe the circumstances.” *Acceptance*: no reported access exclusions attributable to the accessory, and zero reports of perceived instability that analysis traces to a real loss of stability margin.

Design reviews DR-4 and DR-5: the full team, clinical advisor, and independent reviewer, presenting verification reports and deviations, the closed traceability matrix, the human factors validation report, use-study results, and the residual risk assessment. *Exit criteria*: all design inputs verified; all user needs validated under actual or simulated use conditions; residual risks judged acceptable against benefit; no unresolved use error of harmful severity.

Phase 6 — Design Transfer and Release (Weeks 63-72, overlapping)

Process validation (IQ/OQ/PQ) on canopy sewing and rib forming, operator training, first-article inspection, final DHF review and design release signature, then commercial launch at Week 72.

Schedule, Milestones, and Critical Path

Phase	Weeks	Duration	Milestone / review
0 — User Needs	1-8	8 wk	M1: user needs baselined — DR-1
1 — Design Input	9-18	10 wk	M2: PDS and requirements baselined — DR-2
2 — Design Process	19-34	16 wk	M3: concept selected (Wk 26), design freeze (Wk 34) — DR-2b
3 — Design Output	31-42	12 wk	M4: outputs released, protocols approved — DR-3
4 — Verification	43-54	12 wk	M5: verification complete — DR-4
5 — Validation	55-68	14 wk	M6: validation complete — DR-5
6 — Design Transfer	63-72	10 wk	M7: design release and commercial launch

Total duration is roughly 72 weeks, or 16 to 17 months. The **critical path** runs: user needs → wind-load and stability analysis (TA-1, which sets the auto-stow requirement and therefore the actuator specification) → actuator procurement (12-14 week lead) → output release → the 10,000-cycle durability run, which is roughly six weeks of continuous testing and cannot be compressed → validation. Two items are started deliberately early because they are long-lead and easy to underestimate: injection tooling for the rib and hub components (12-16 weeks, released at M4), and IRB approval for the home-use study, submitted during Phase 3 rather than Phase 5 because an 8-10 week review would otherwise land directly on the critical path. Roughly four weeks of float is carried in Phase 2 to absorb a failed concept screen, which is the most likely single source of schedule loss.

References

- [1] U.S. Food and Drug Administration, *Design Control Guidance for Medical Device Manufacturers*. Rockville, MD: Center for Devices and Radiological Health, March 11, 1997.
- [2] *Code of Federal Regulations*, Title 21, Part 820.30 — Design Controls. U.S. Government Publishing Office. (Part 820 has been amended by FDA's Quality Management System Regulation, which incorporates ISO 13485:2016 by reference; design control requirements map to ISO 13485 clause 7.3.)
- [3] G. E. Dieter and L. C. Schmid, *Engineering Design*, 5th ed. New York: McGraw-Hill, 2013.
- [4] R. A. Cooper, *An Introduction to Rehabilitation Engineering*. Boca Raton, FL: Taylor & Francis, 2007.
- [5] International Organization for Standardization, ISO 7176 series, *Wheelchairs* — in particular Part 1 (determination of static stability), Part 2 (determination of dynamic stability of electric wheelchairs), Part 4 (energy consumption and theoretical distance range), Part 5 (determination of dimensions, mass and manoeuvring space), Part 8 (requirements and test methods for static, impact and fatigue strengths), Part 9 (climatic tests), Part 14 (power and control systems), Part 19 (wheeled mobility devices for use as seats in motor vehicles), and Part 21 (electromagnetic compatibility). Geneva: ISO.
- [6] U.S. Department of Justice, *2010 ADA Standards for Accessible Design*, §307.4 (protruding objects) and §309.4 (operable parts). Washington, DC, 2010.
- [7] International Electrotechnical Commission, *IEC Guide 71: Guidelines for Standards Developers to Address the Needs of Older Persons and Persons with Disabilities*. Geneva: IEC, 2001.
- [8] International Organization for Standardization, *ISO 14971: Medical Devices — Application of Risk Management to Medical Devices*. Geneva: ISO.
- [9] U.S. Food and Drug Administration, *Applying Human Factors and Usability Engineering to Medical Devices: Guidance for Industry and Food and Drug Administration Staff*. February 3, 2016.
- [10] L. Demers, R. Weiss-Lambrou, and B. Ska, "The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress," *Technology and Disability*, vol. 14, no. 3, pp. 101-105, 2002.
- [11] U.S. Food and Drug Administration, *Deciding When to Submit a 510(k) for a Change to an Existing Device: Guidance for Industry and Food and Drug Administration Staff*. October 25, 2017.